# AUDIO TECHNOLOGY 2 UE02

# FREQUENCY CORRECTION

goal: linearer frequency response, adapted to a specific scenario (eg. studio, club, live venue)

signal generator → loudspeaker → measuring mic → analyzer

graphic or parametric EQ is set on DSP, mixing console or in DAW


frequency response varies depending on the position of the loudspeaker and the measuring mic

- comb filter effects (superposittion of identical, delayed signal eg. through reflexions off the ground)
- room modes (standing wave between parallel walls)

# MEASURING MICROPHONE


pressure receiver, omnidirectional, linear down to 10Hz, electret capsule, can also be used for recordings

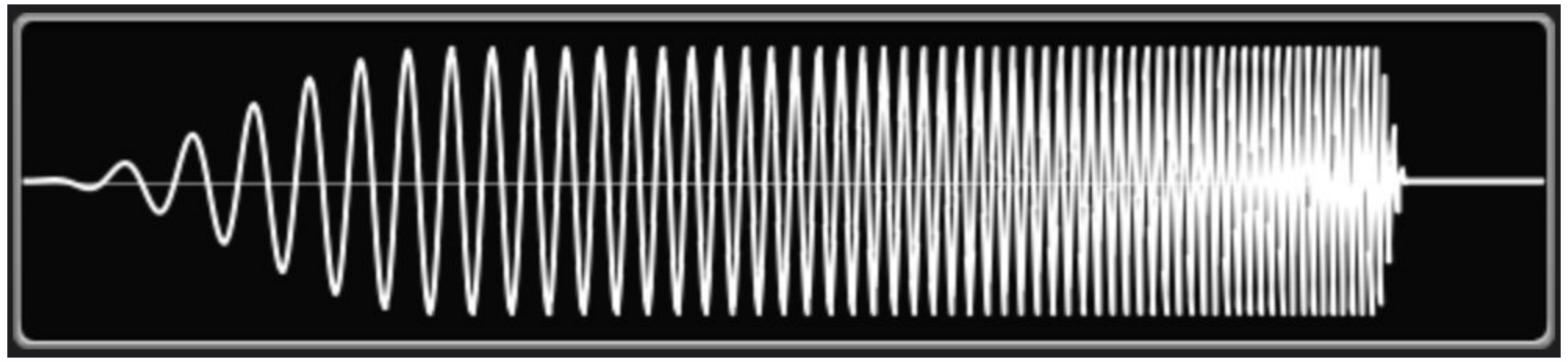
use either high quality microphones ot use a correction filter provided by the manufacturer

zB.: earthworks, UMIK-1

# PINK NOISE VS SINE SWEEP

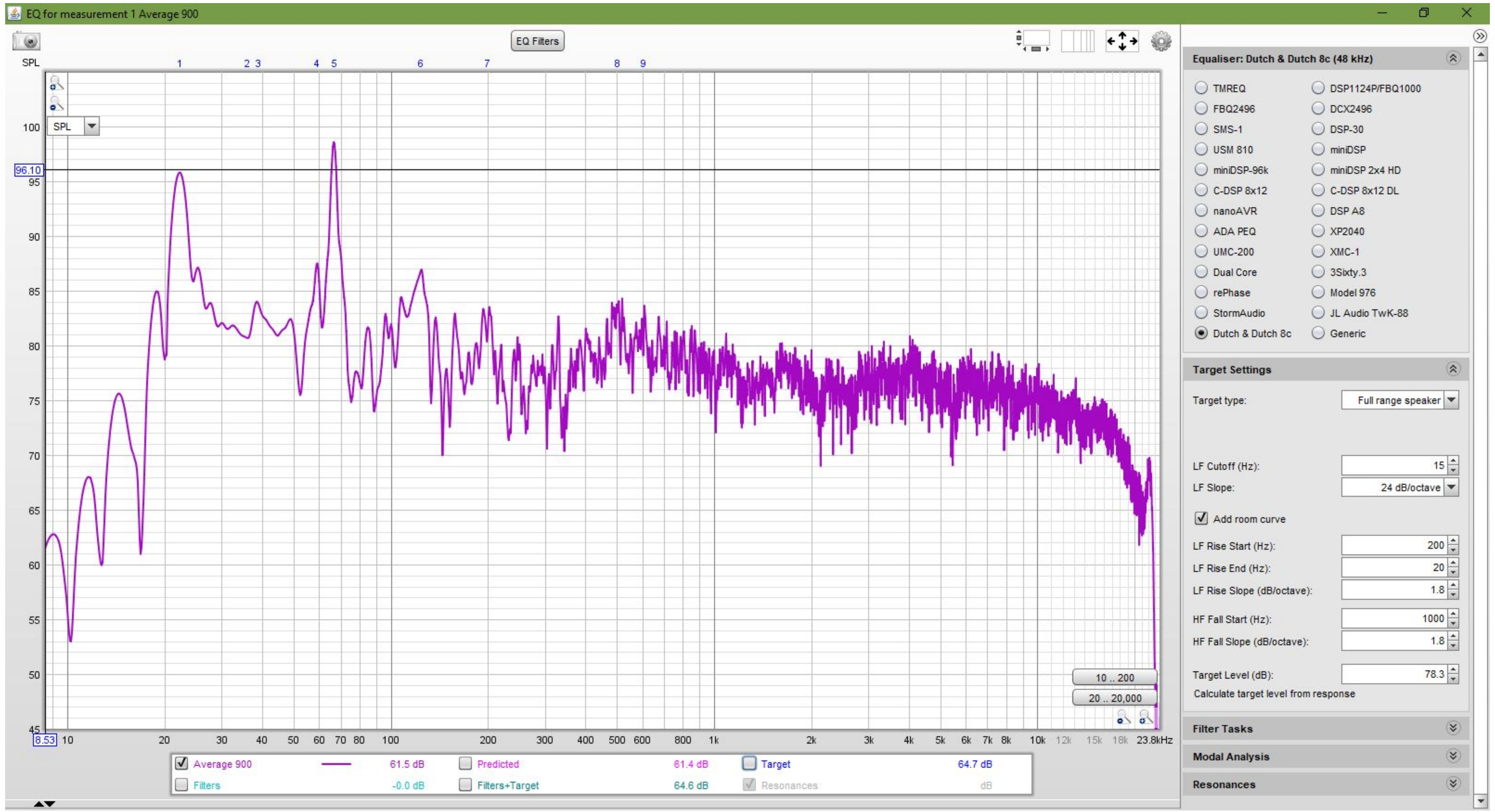
pink noise gives you an idea about the sound, can reveal harsh frequencies
frequencies cannot be generated all at once - random or periodically changing
allows real time correction and measurements

sine sweep puts out more energy/frequency
better SNR (even better for longer sweeps)
higher precision



# TASK: MEASURING A STUDIO MONITOR IN REW

- install REW
- load calibration file from manufacturer
- run a measurement using a sine sweep
- find an EQ and apply it
- measure again
- Listen to music with and without correction!



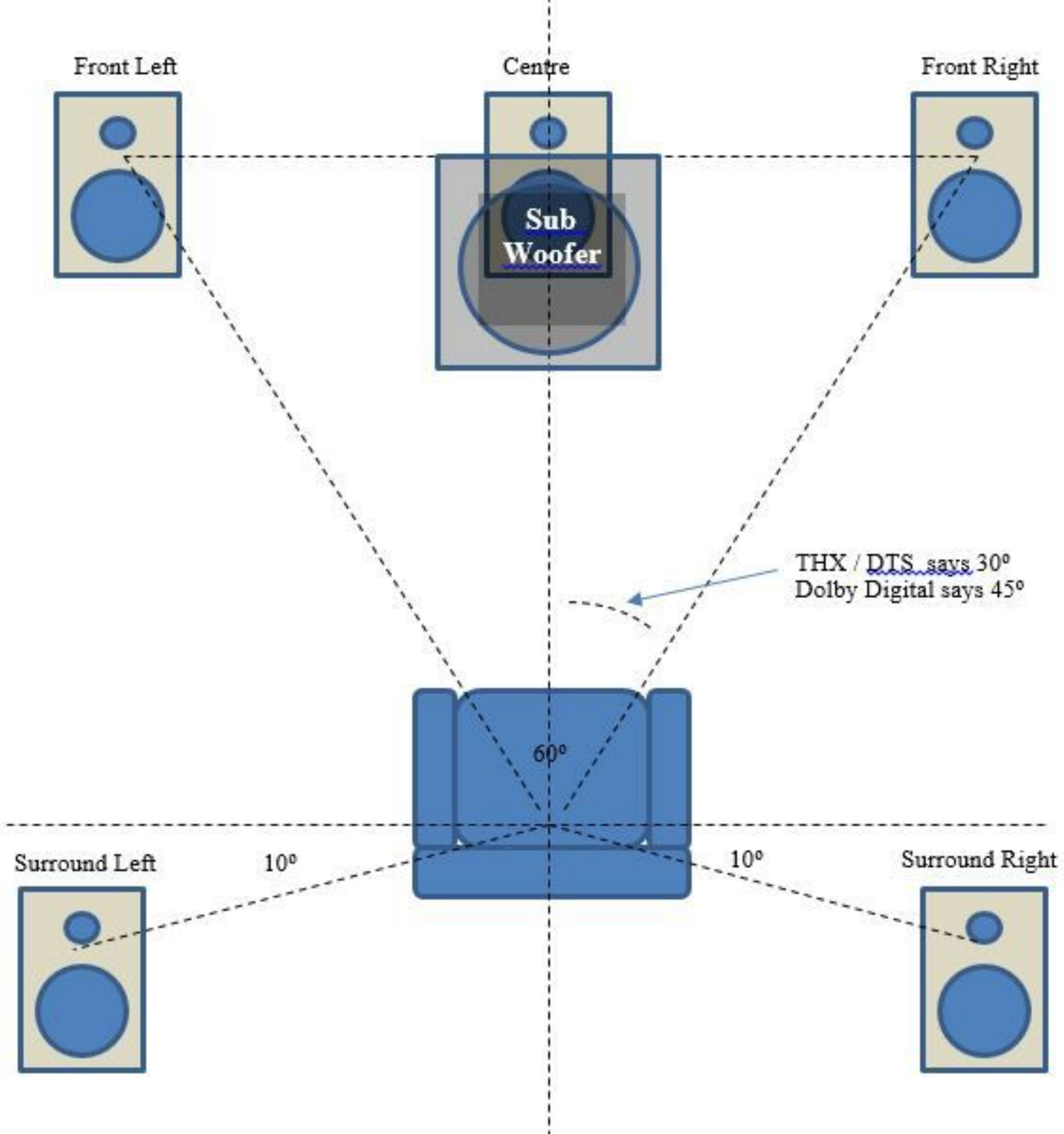

How much variation is there between the measurements?
How does the correction affect your perception of the music?

# Spatial audio

# How can we capture it?


(a) (b) (c)

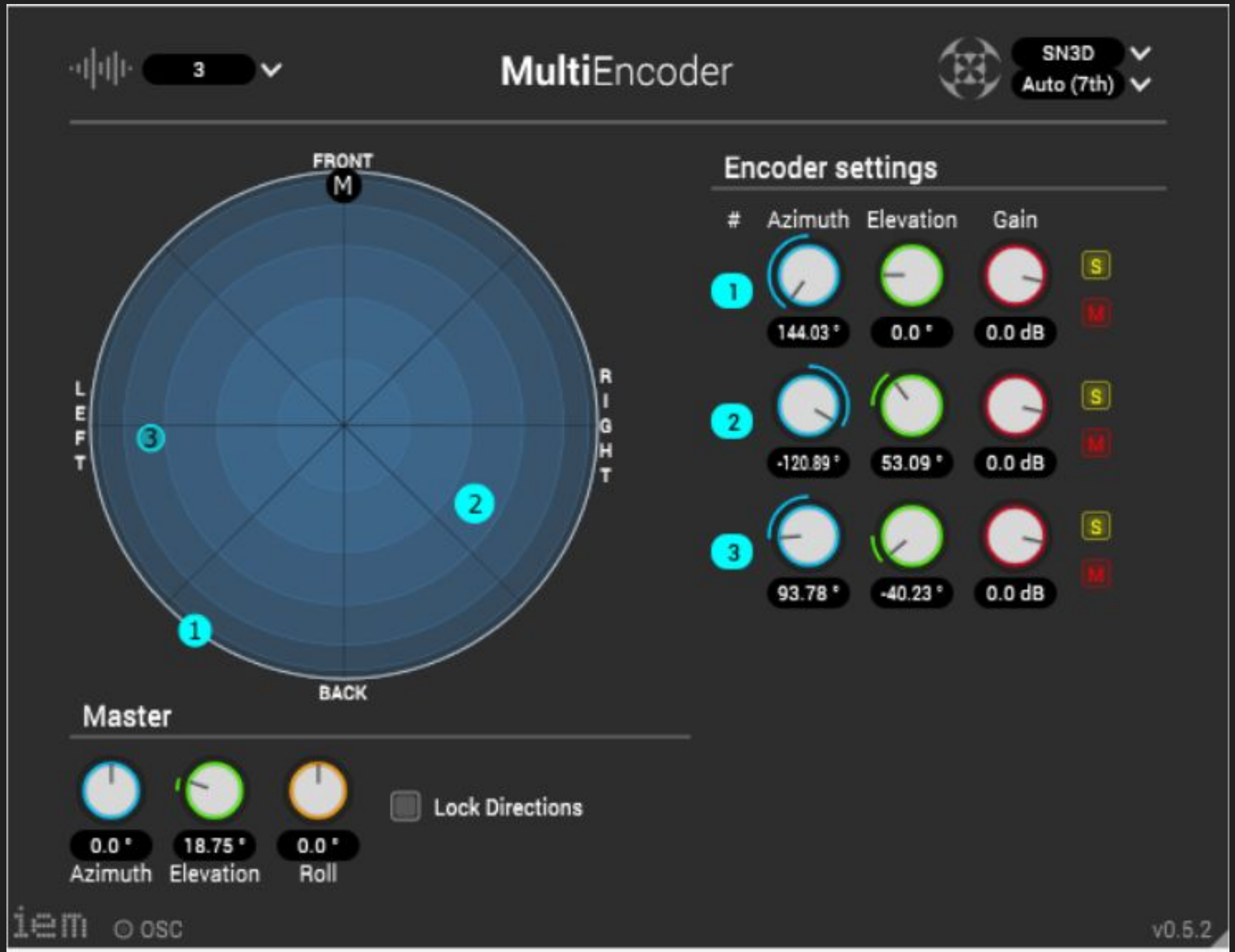
How can we reproduce it?


Front Left
Centre
Front Right
Sub
Woofer
THX / DTS says 30º
Dolby Digital says 45º
60º
Surround Left
10º
10º
Surround Right

# Dealing with Spatial Audio: Approaches

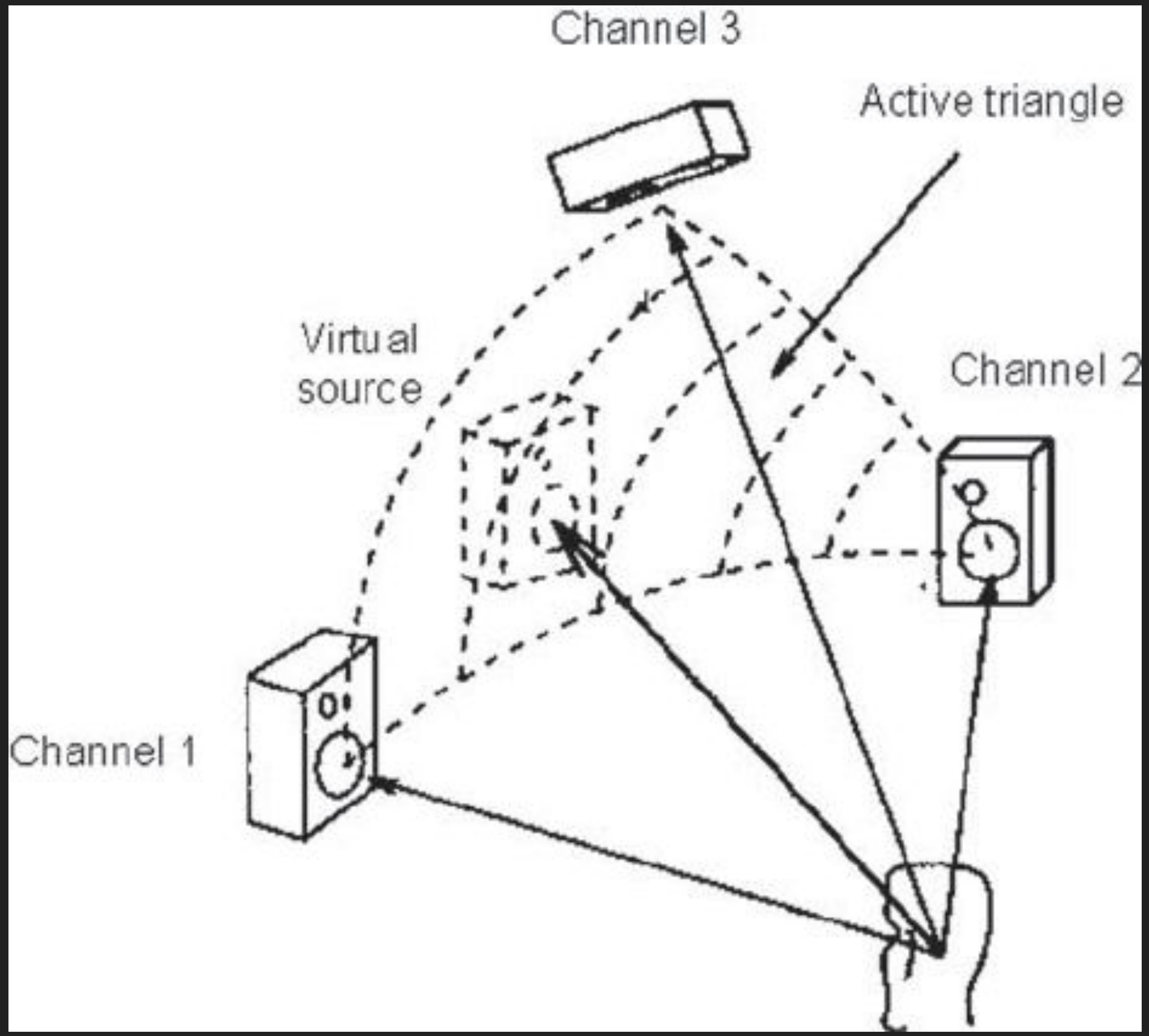
➔ **Channel Based**
A channel-based mix, usually for a specific target loudspeaker

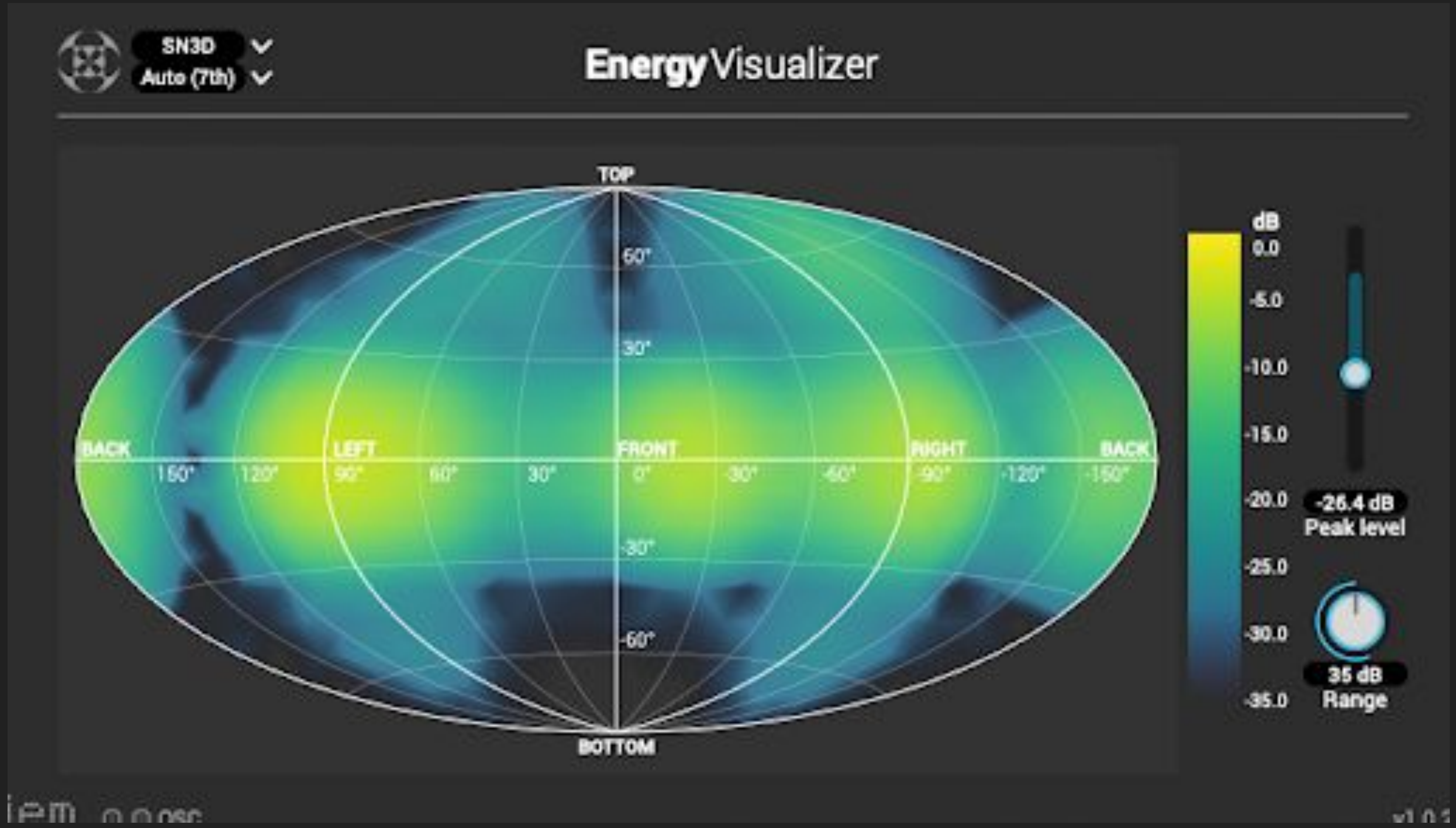
➔ **Object Based**
Encoding each audio source independently with positional metadata, and letting the renderer at the reproduction side position the audio as best it can to the desired location.

# Modern Spatial Audio Renderers (Panners)

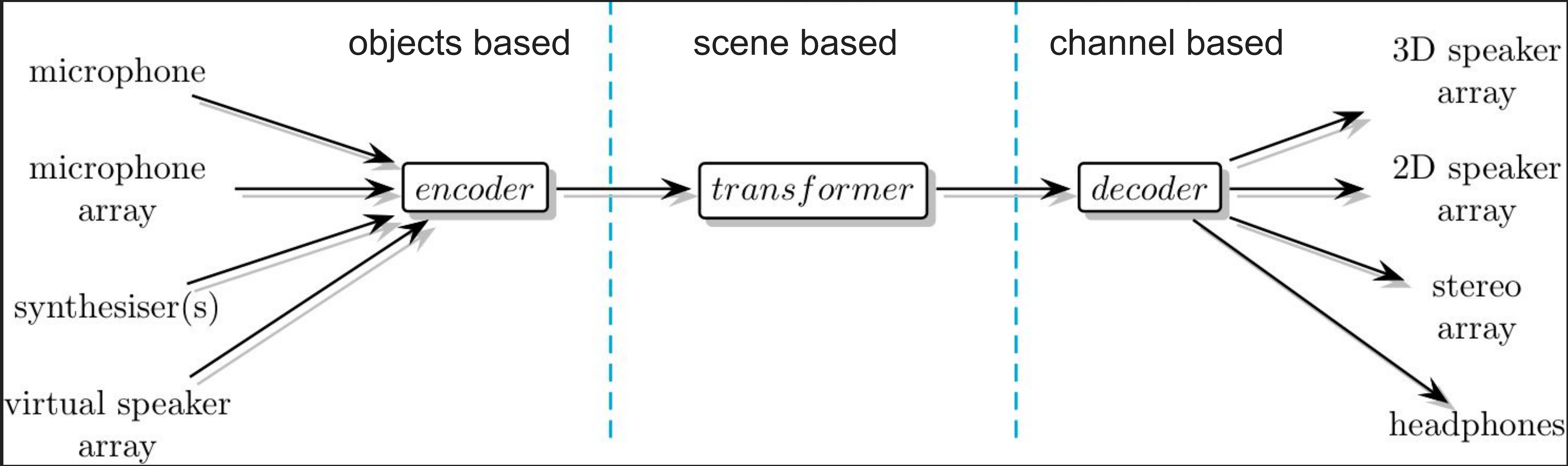
- ➔ **VBAP** (Vector based amplitude panning)
  Interaural level difference (ILD) based on triangles. Only three speakers are used at maximum.
- ➔ **Ambisonic Panning**
  The audio scene is modelled as a sphere (non-preferred direction). There is always sound played in all speakers.



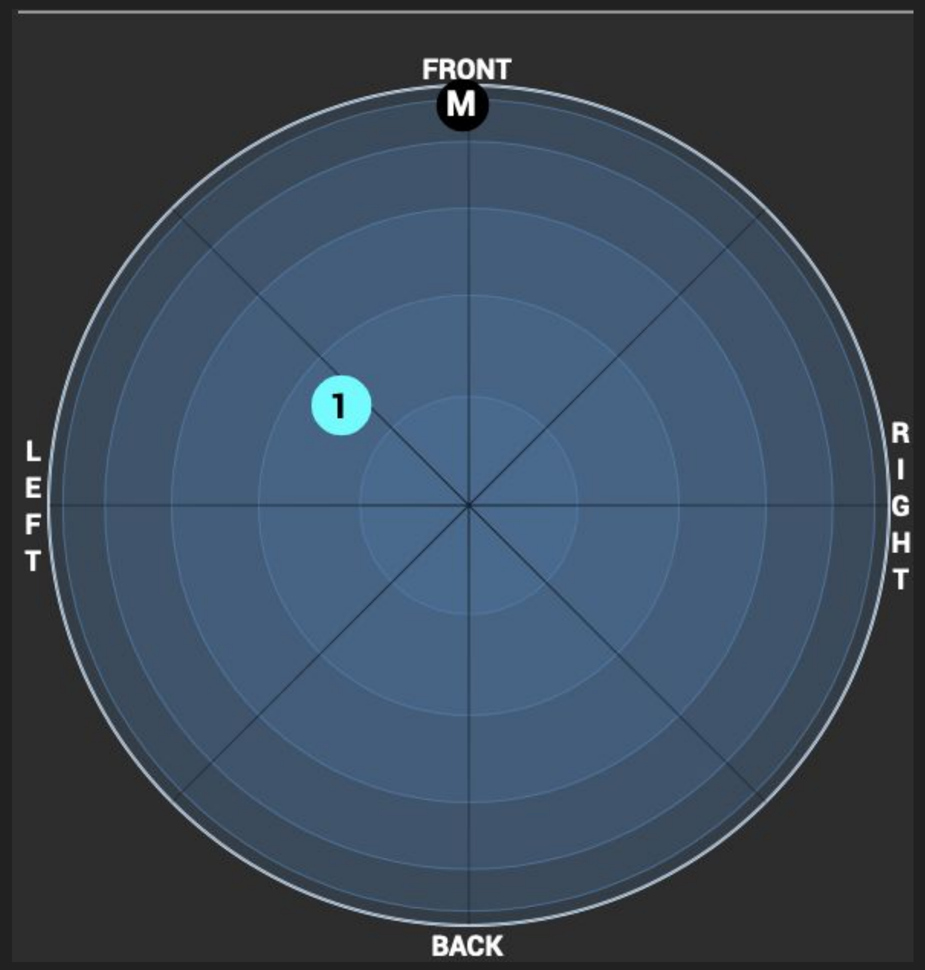



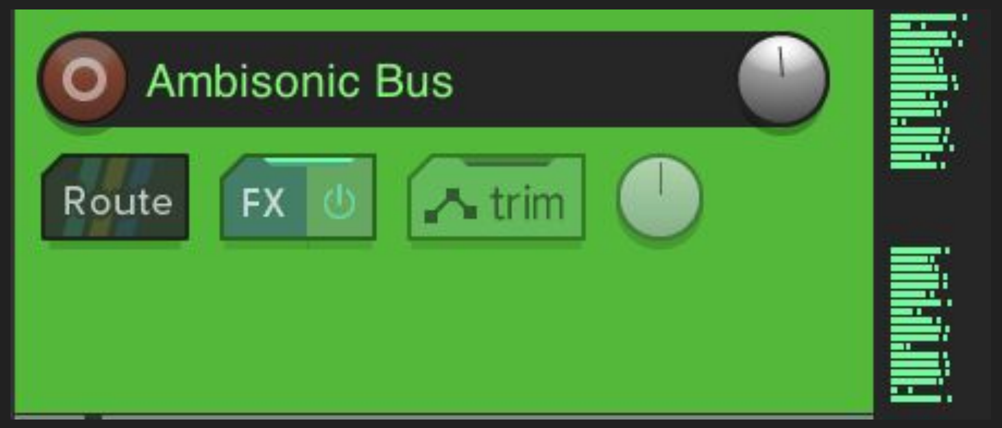
# Ambisonics Workflow

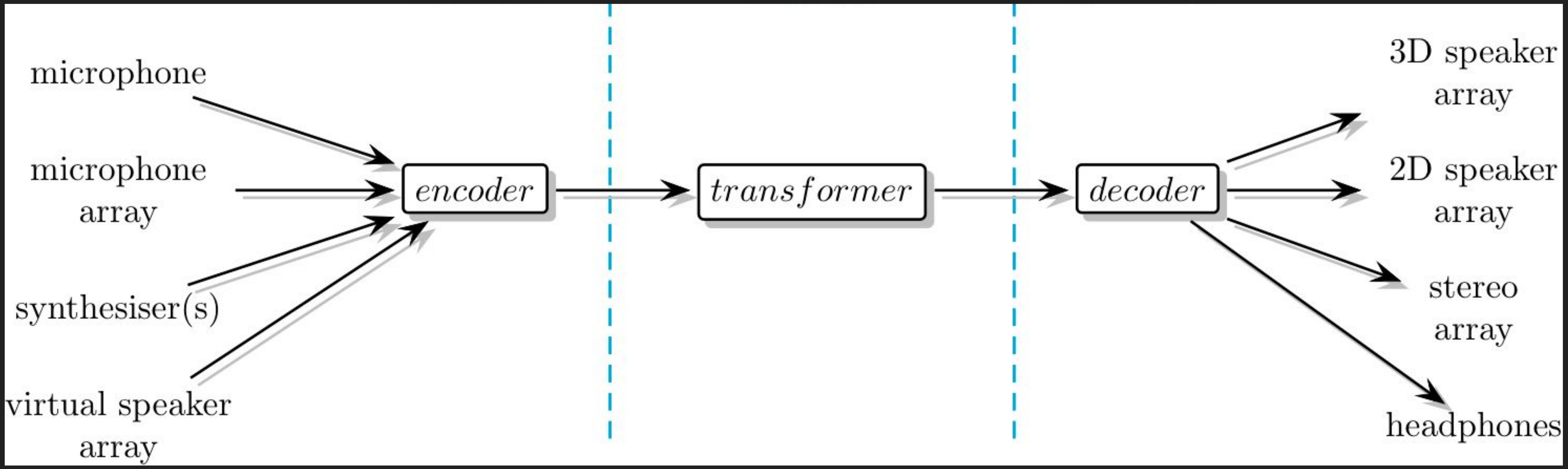
Audio contents are first encoded in the ambisonic audio format and later decoded to a particular speaker layout



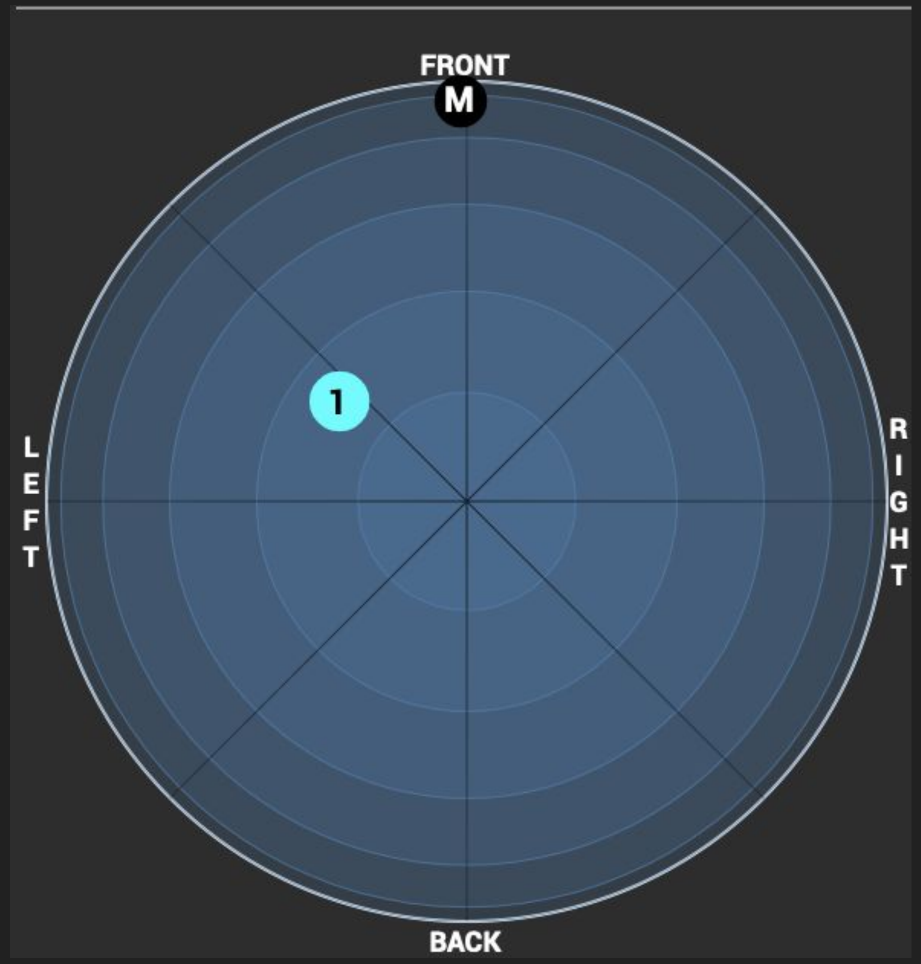
The ambisonics encoded audio is the decomposition of an audio signal into a number of spherical harmonics. For example, here a mono track is encoded into a new 25 channels audio signal using 4th order ambisonics.

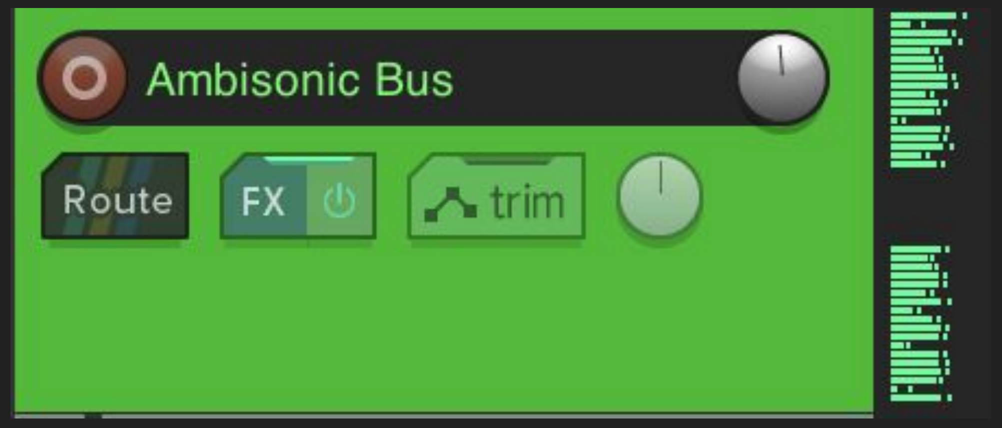



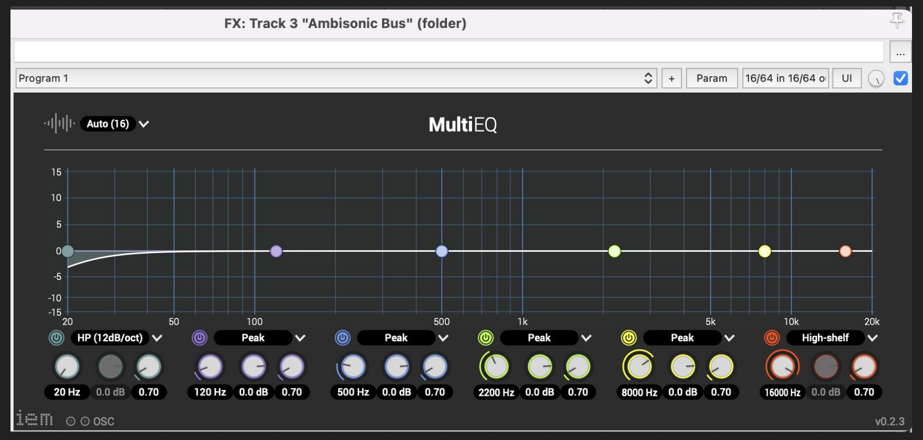



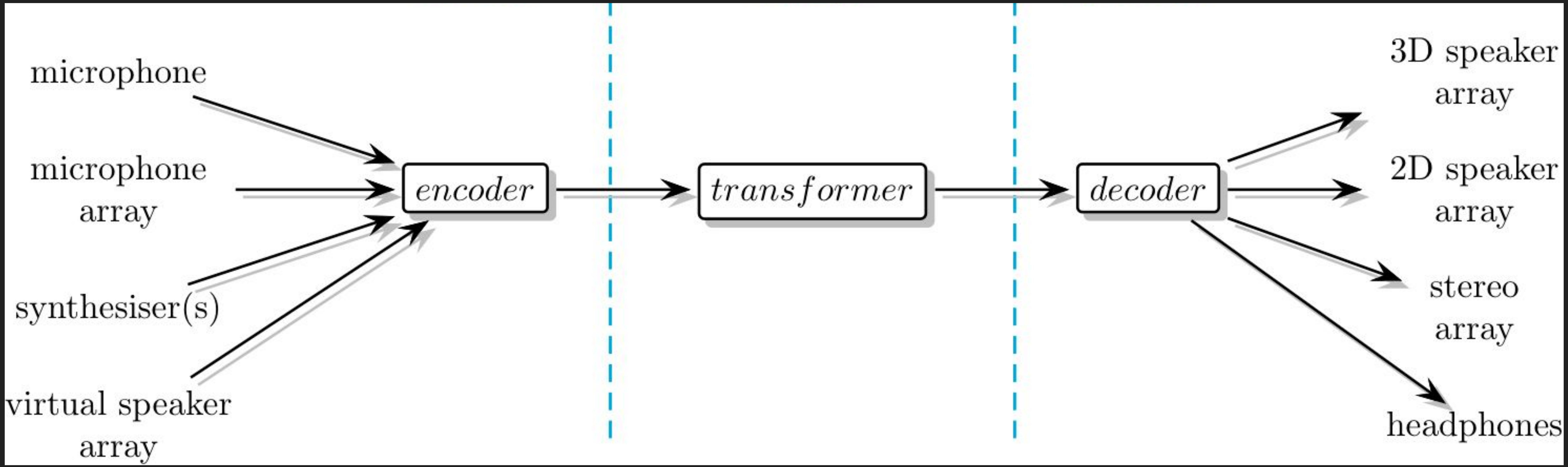



# The transformer can be any ambisonics effect: reverbs, EQ, rotations, spread effects, room effects, etc

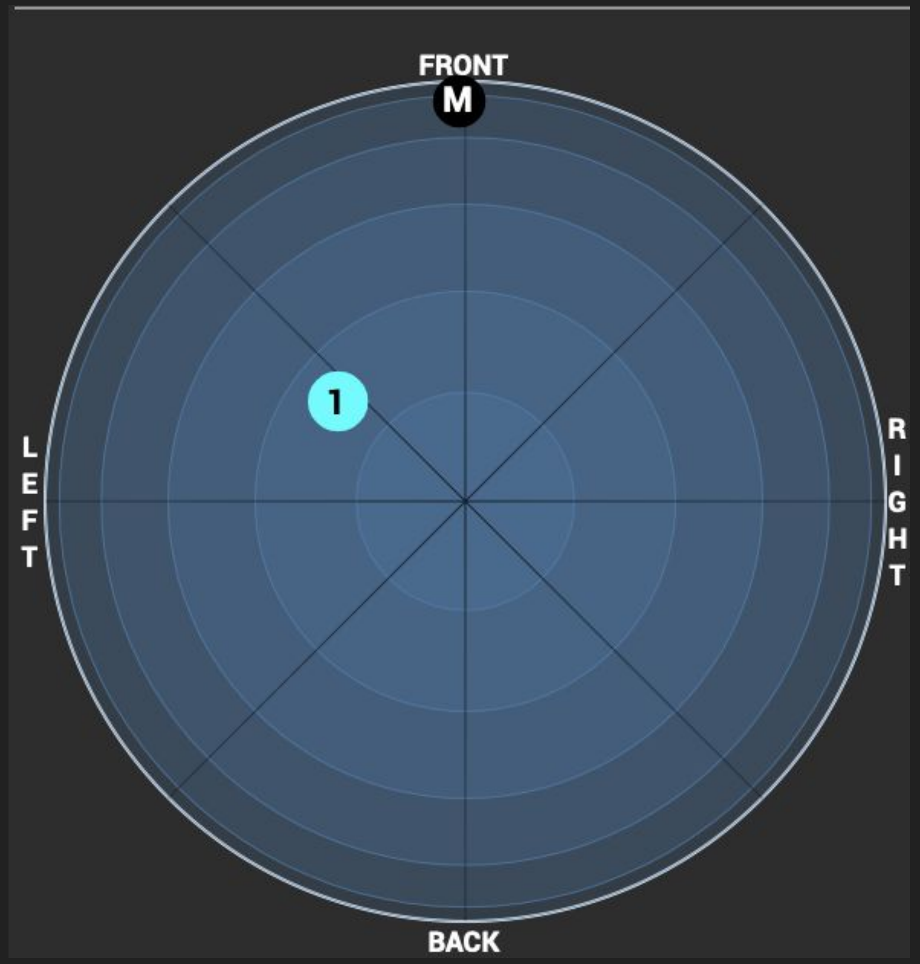



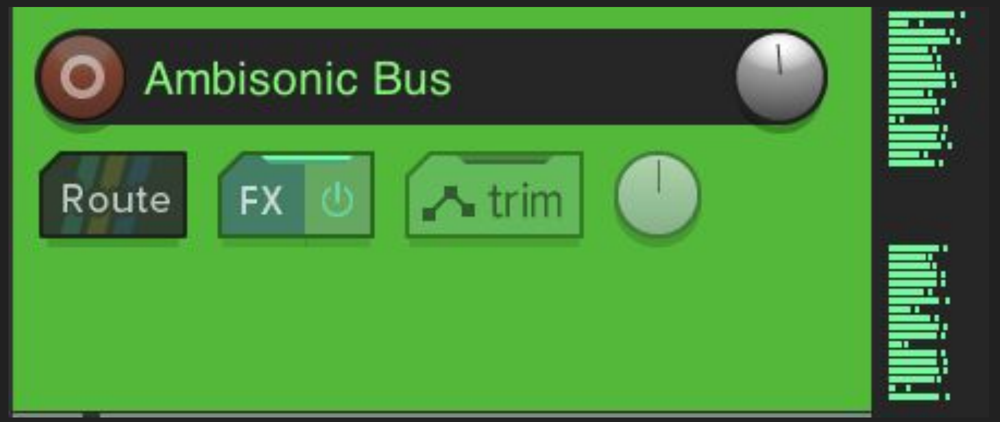



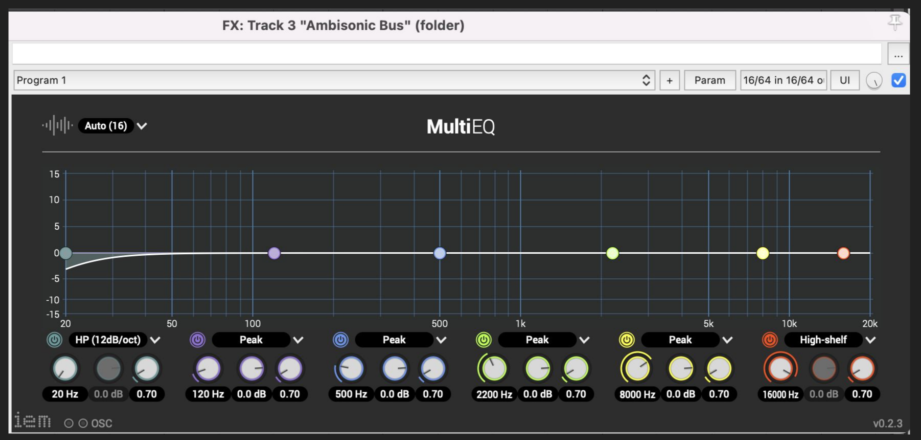



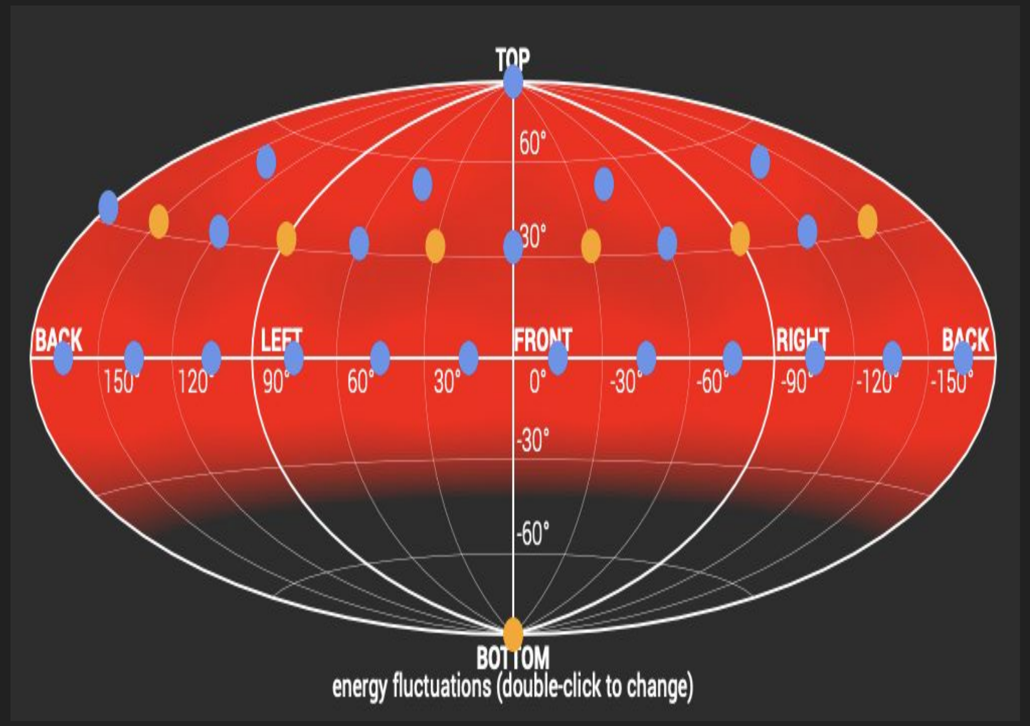



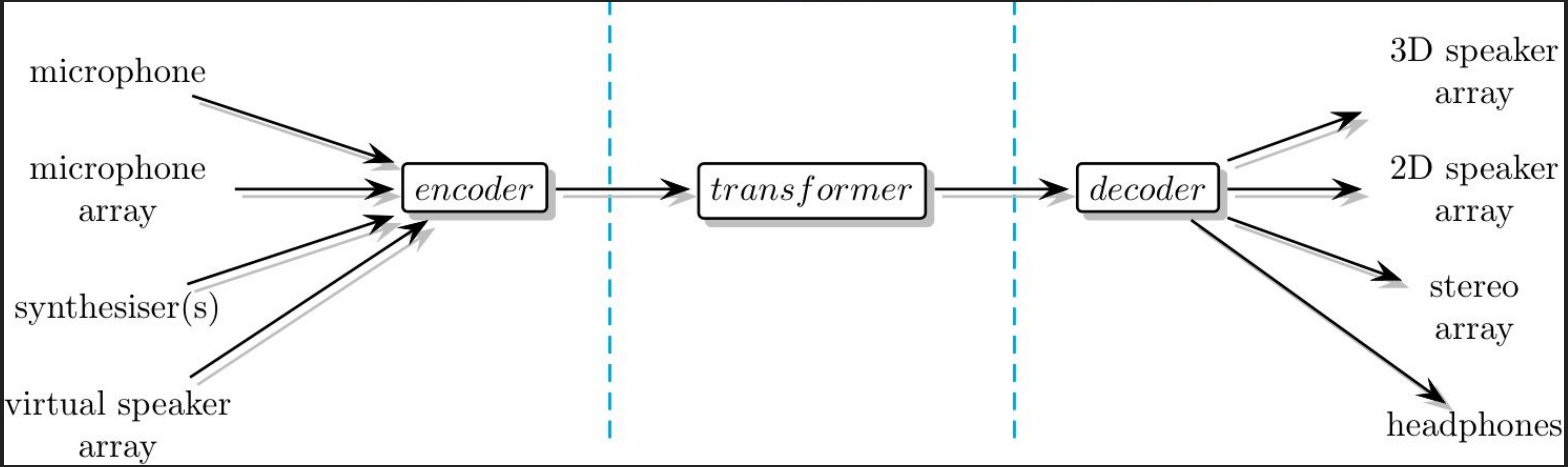
The number of audio channels of the encoded signal usually does NOT coincide with the number of speakers used (output channels). It depends on the ambisonic order uses. Here we add a decoder for rendering the output audio channels to be played at the actual speaker layout.

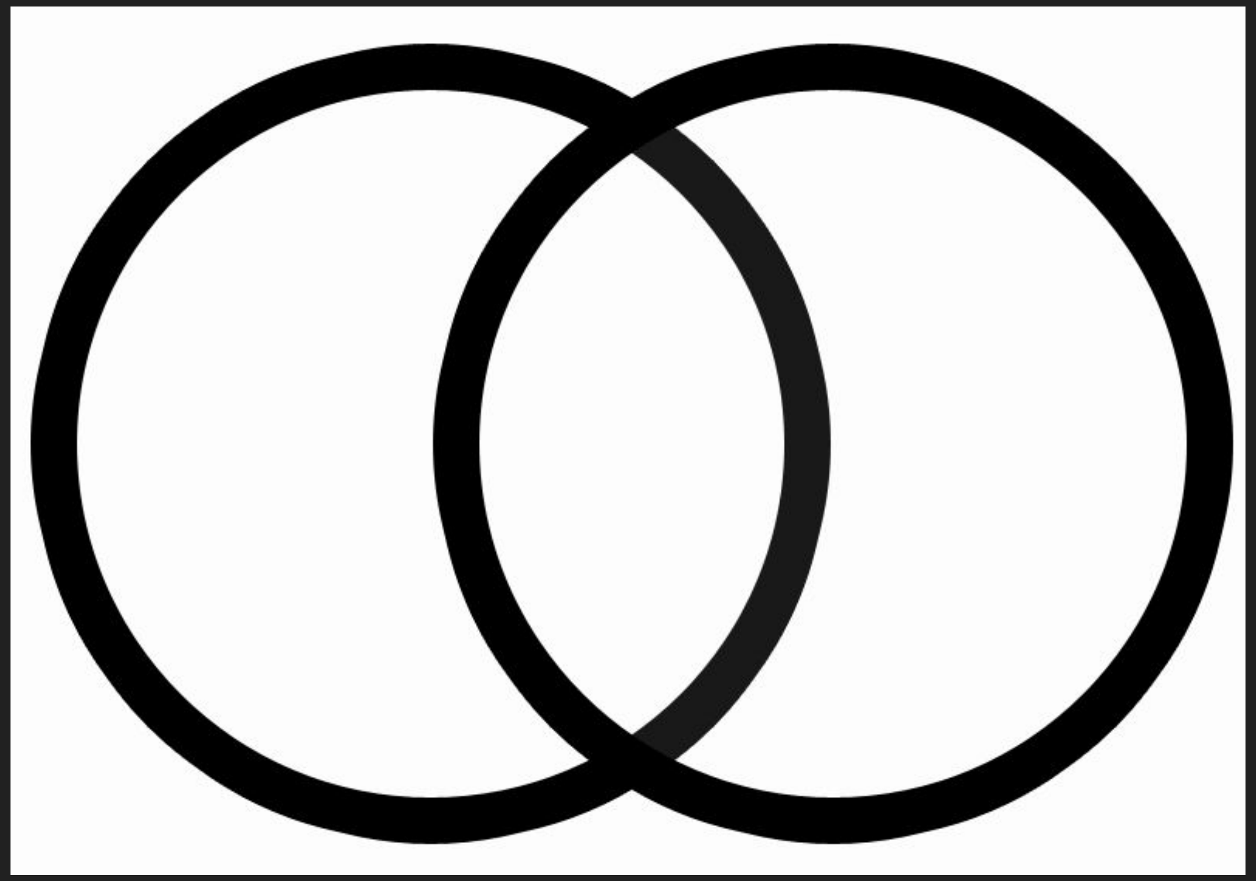



# Spatial Orders

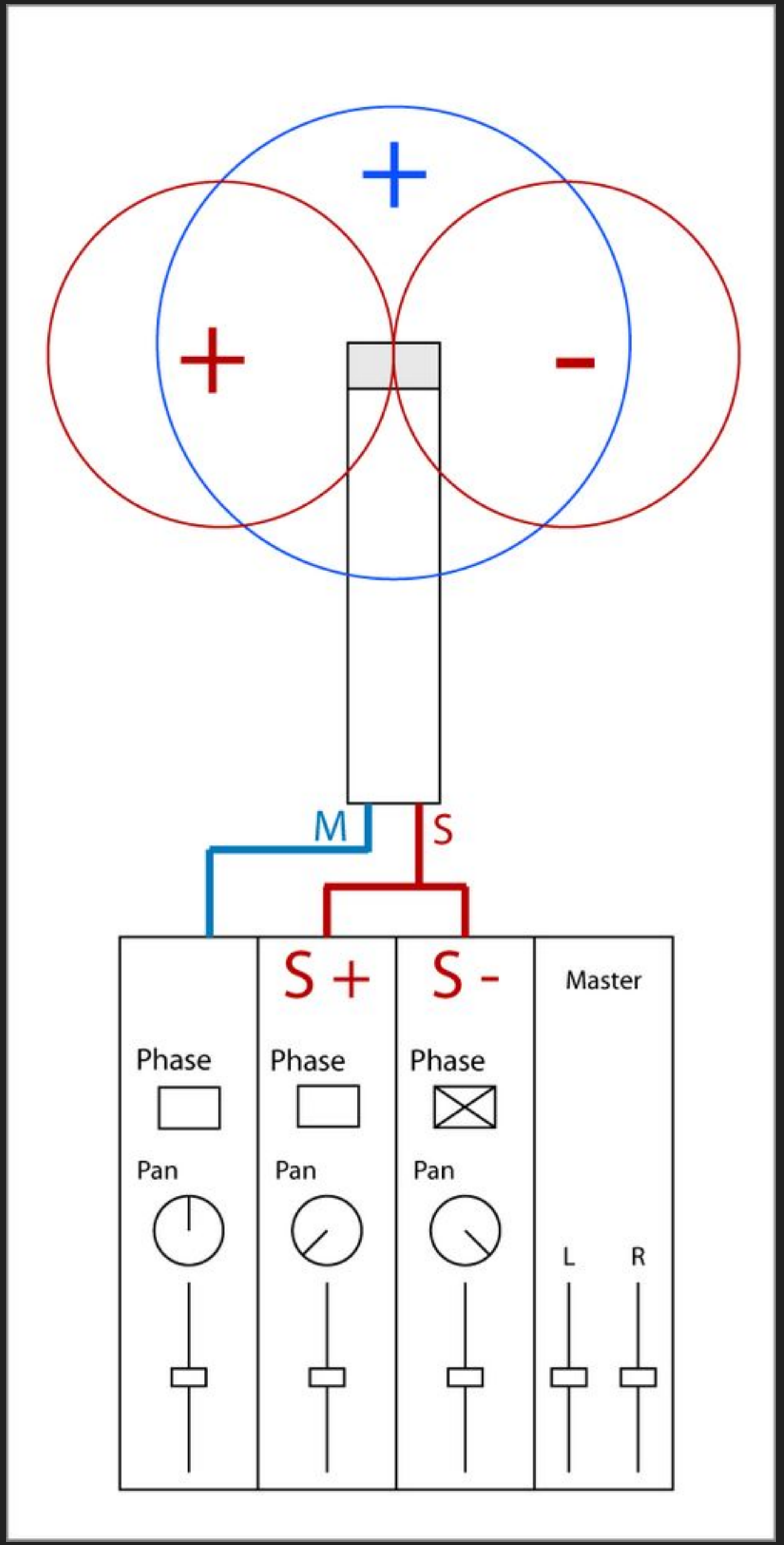
Extending the M-S format



stereo: left and right channel that overlap

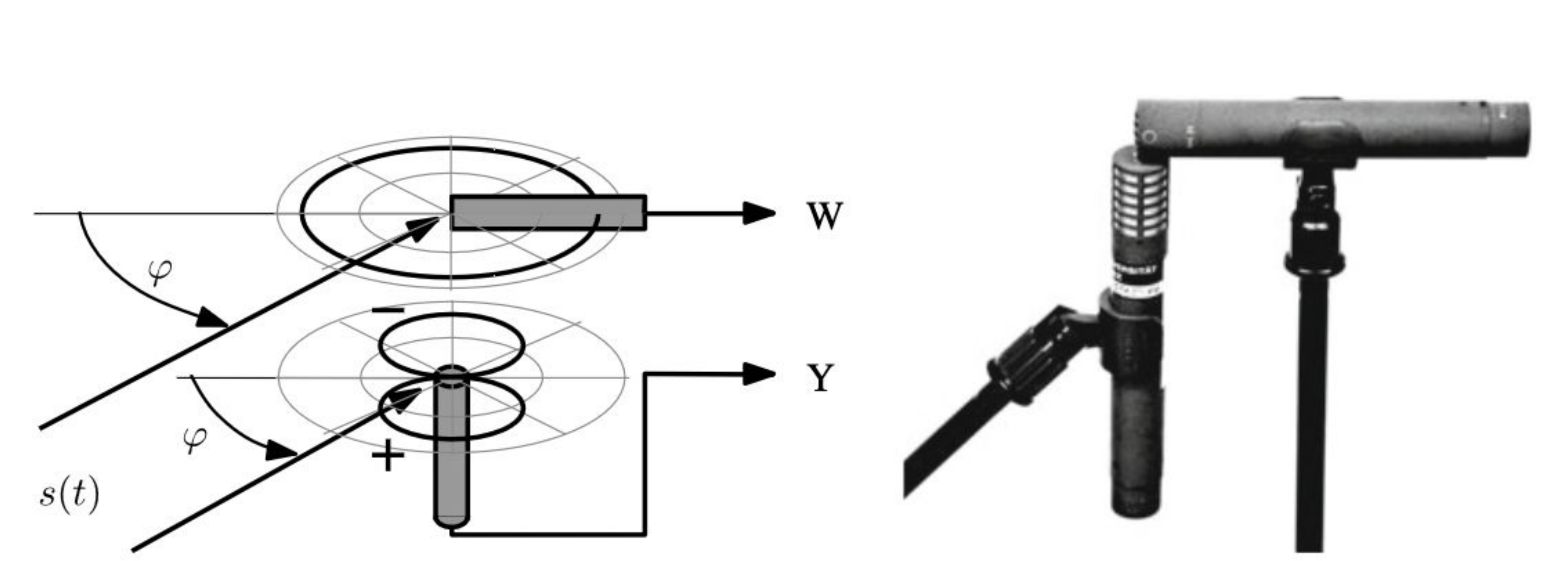
$$M = L + R$$
$$S = L - R$$

$$L = M + S$$
$$R = M - S$$



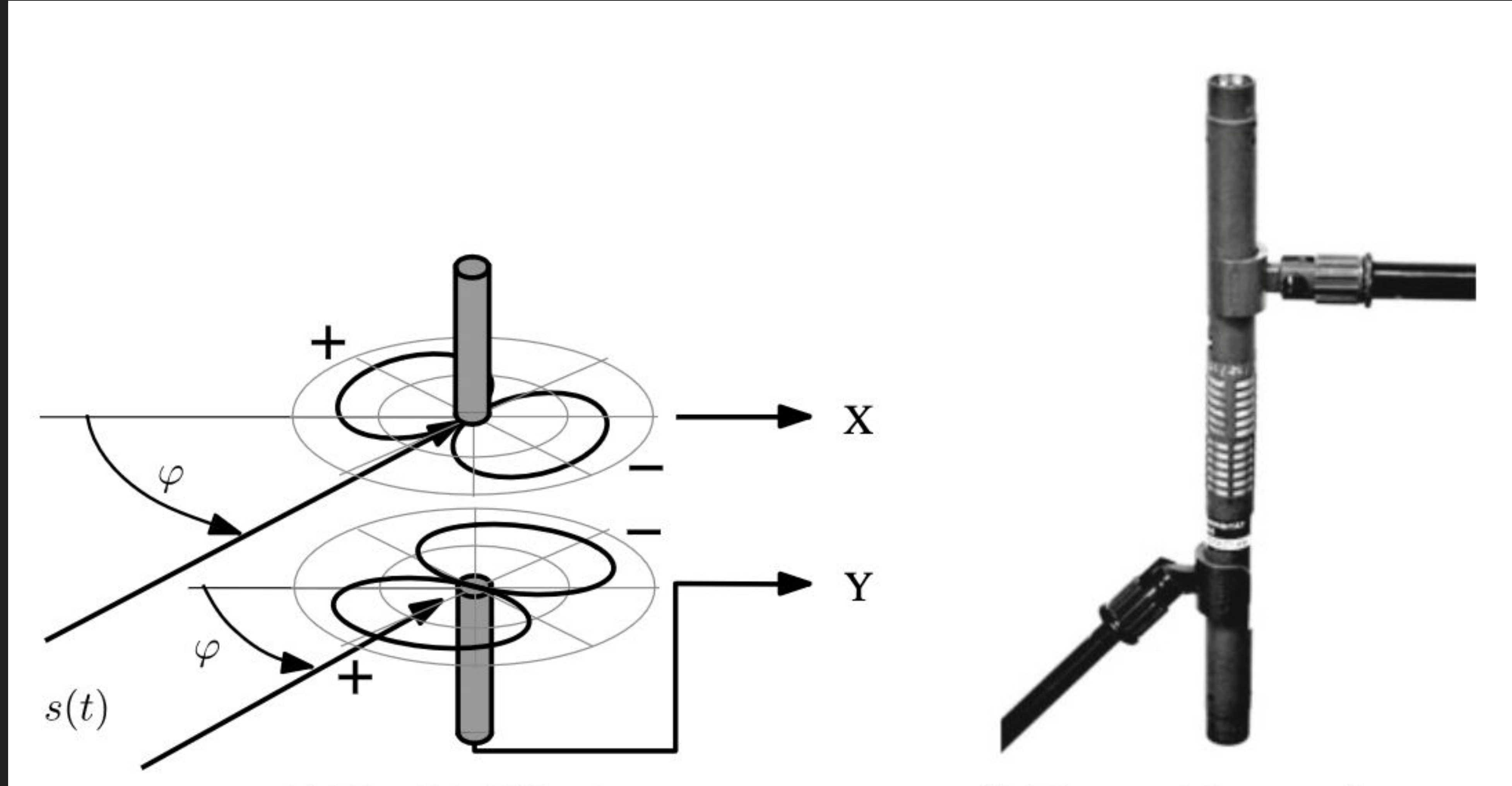
# Spatial Orders

## Extending the M-S format



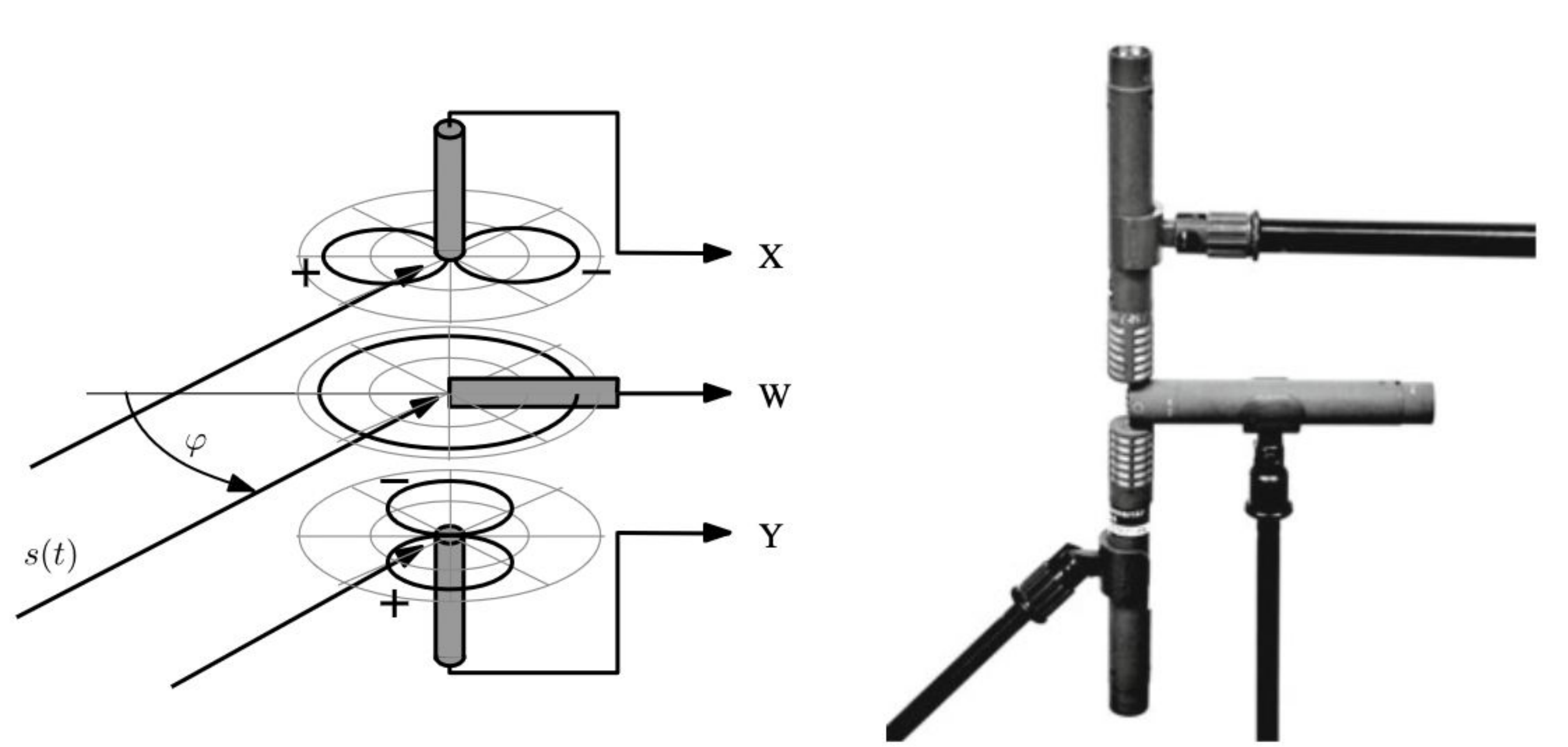

(a) Native MS recording (b) Picture of the recording setup

**Fig. 1.2** Native mid-side recording with the coincident arrangement of an omnidirectional microphone heading front and a figure-of-eight microphone heading left



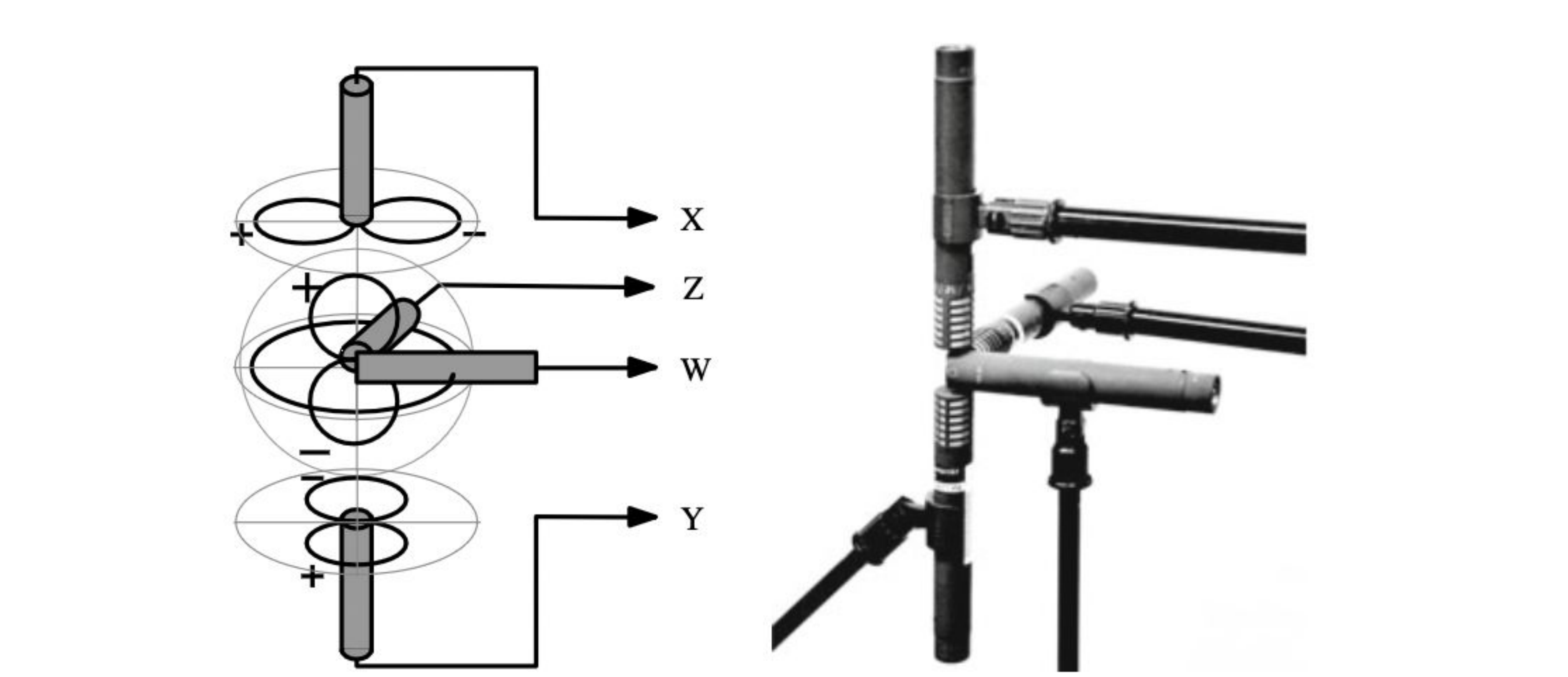

(a) Blumlein XY pair (b) Picture of the recording setup

**Fig. 1.1** Blumlein pair consisting of 90°-angled figure-of-eight microphones




(a) Native 2D FOA recording (b) Picture of recording setup

**Fig. 1.5** Native 2D first-order Ambisonic recording with an omnidirectional and a figure-of-eight microphone heading front, and a figure-of-eight microphone heading left; photo shown on the right

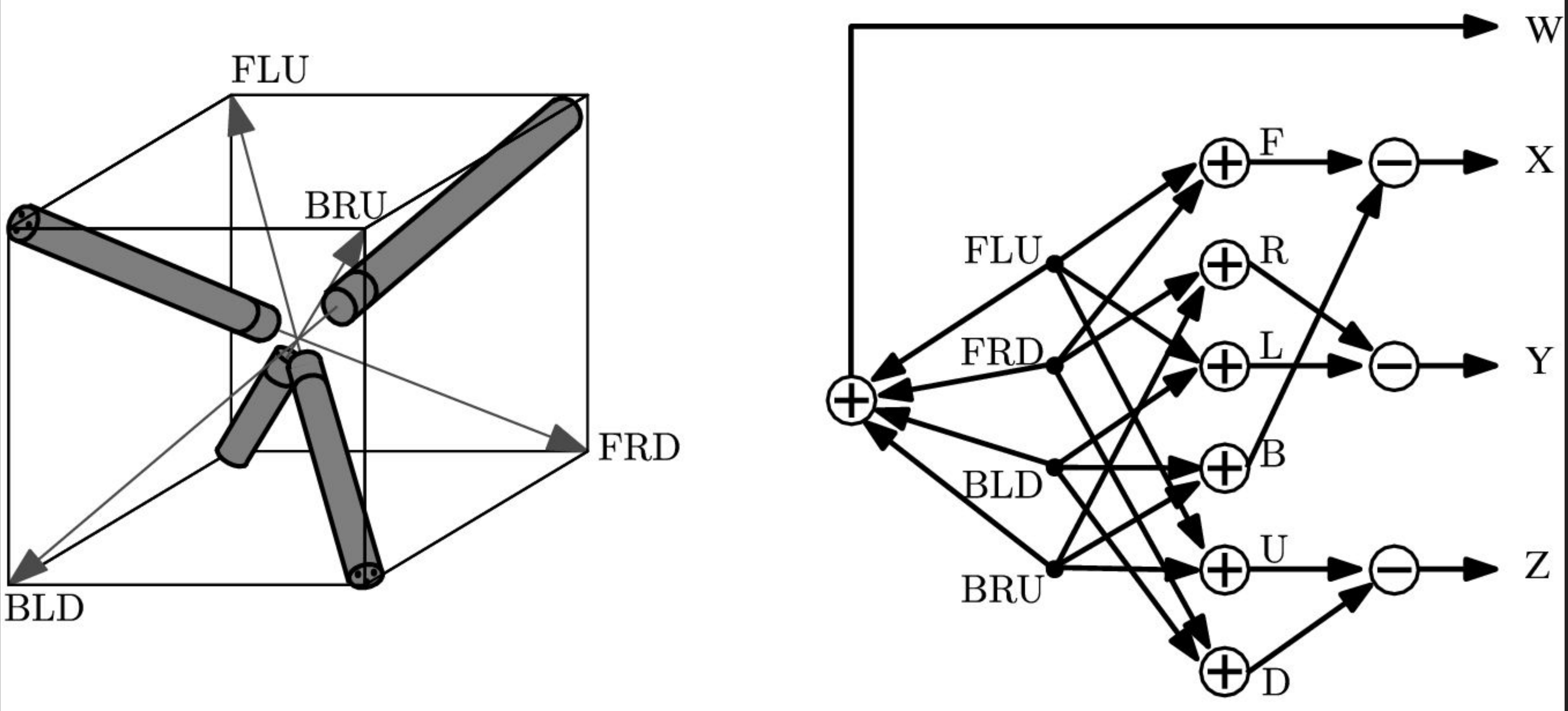



(a) Native 3D FOA recording (b) Picture of the recording setup

**Fig. 1.10** Native 3D first-order Ambisonic recording with an omnidirectional and three figure-of-eight microphones aligned with the Cartesian axes X, Y, Z

# Spatial Orders

Relationships between order and channels




(a) Tetrahedral array with four cardioids (b) Encoder of microphone signals

Result: 4 channels ->three spherical axis X, Y, Z plus an omnidirectional component W

# Spatial Orders

Example First Order: decomposition in three spatial directions

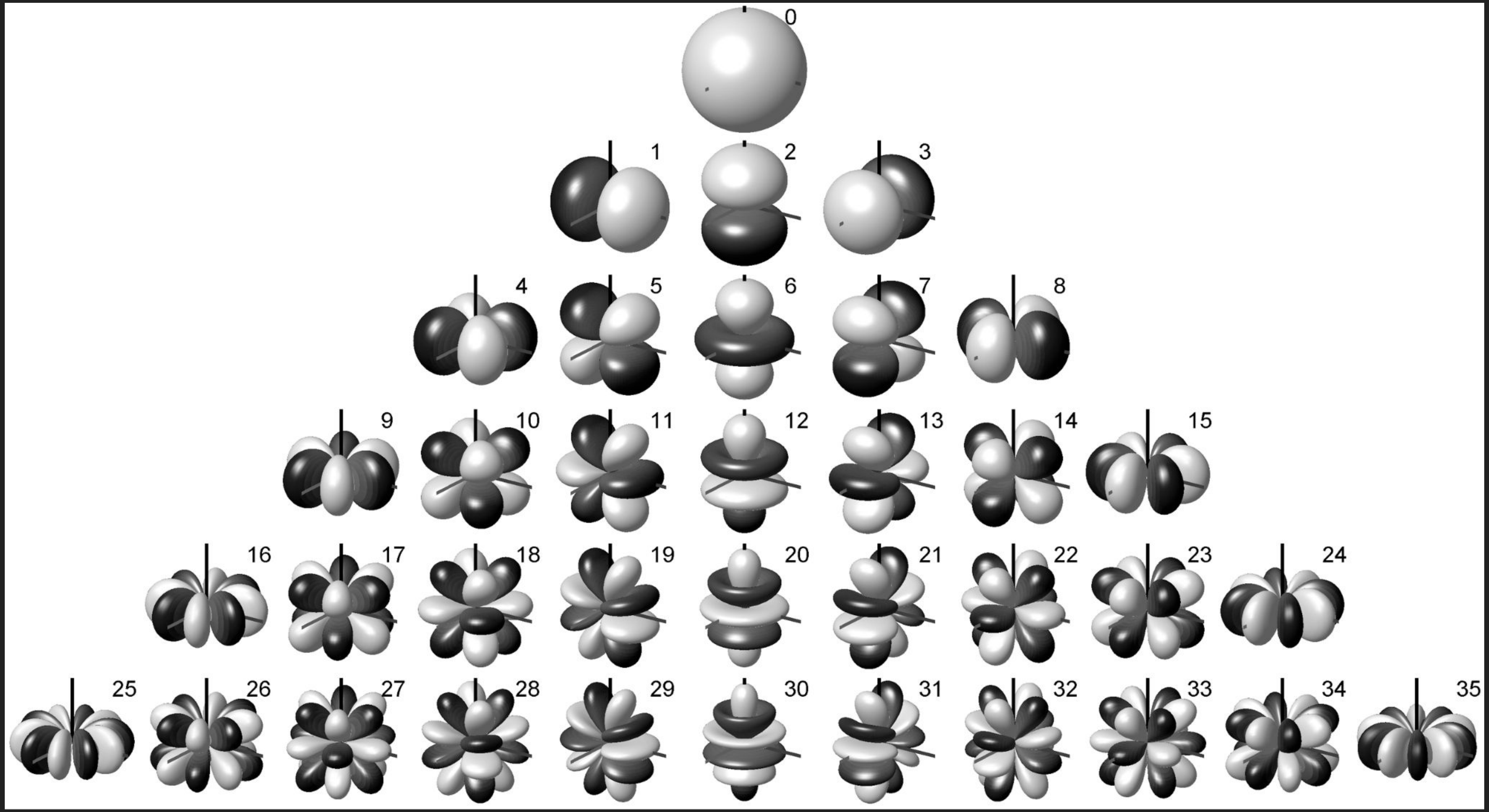
(a) Tetrahedral array of 4 cardioids (b) SPS200 (c) MK4012 (d) ST450

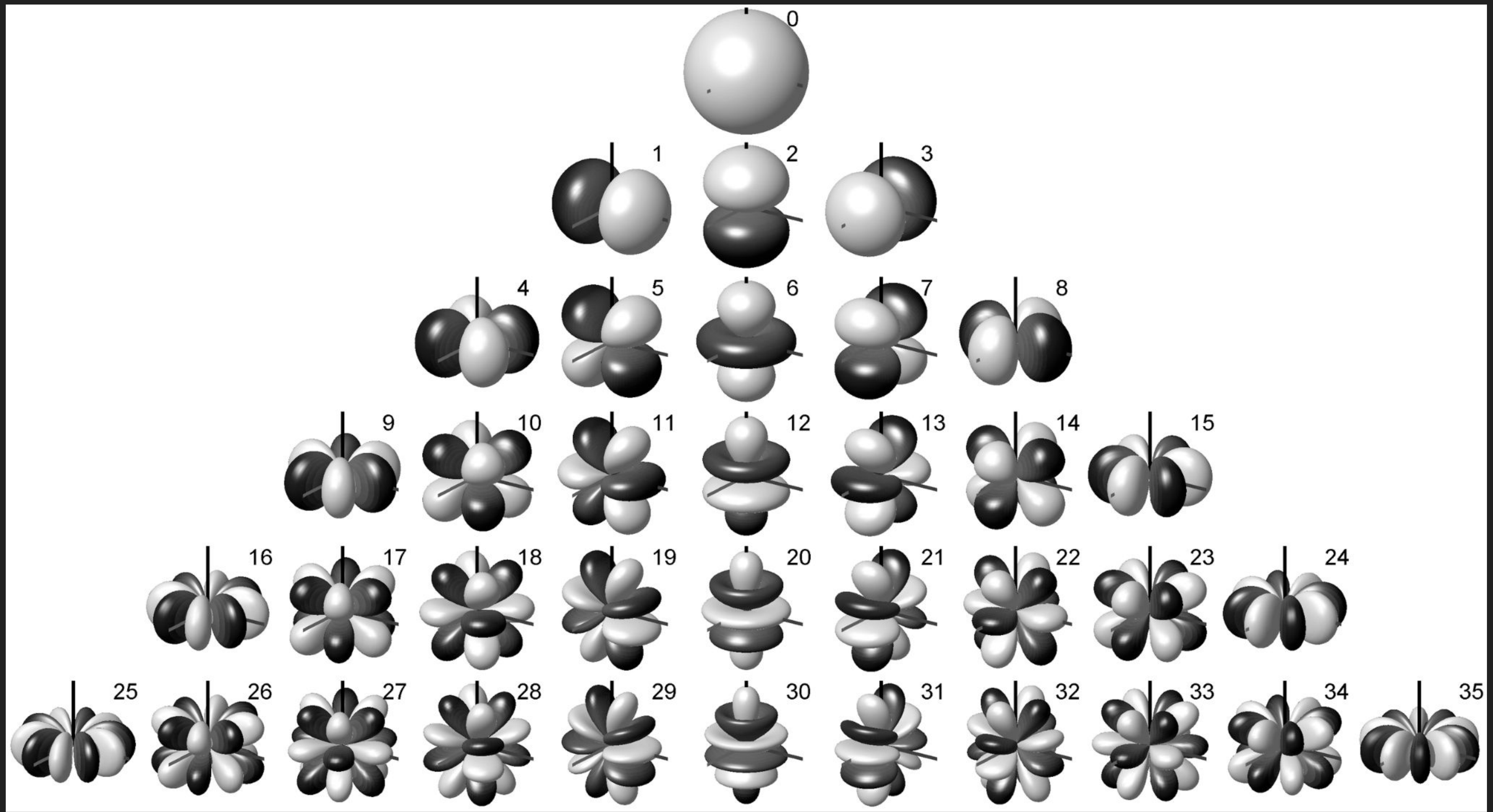
Result: 4 channels ->three spherical axis X, Y, Z plus an omnidirectional component W

# Spatial Orders

Higher Order:

decomposition into more spatial directions (harmonics)






# To remember

➔ **Spatial harmonics**
The more spatial harmonics (higher order), the less "spatially smeared" your soundfield will be.

➔ **Which order?**
We would love to use 100th order but we do not count with pragmatic software or hardware for that. From 4th order or more, the quality of the soundfield improves a lot.

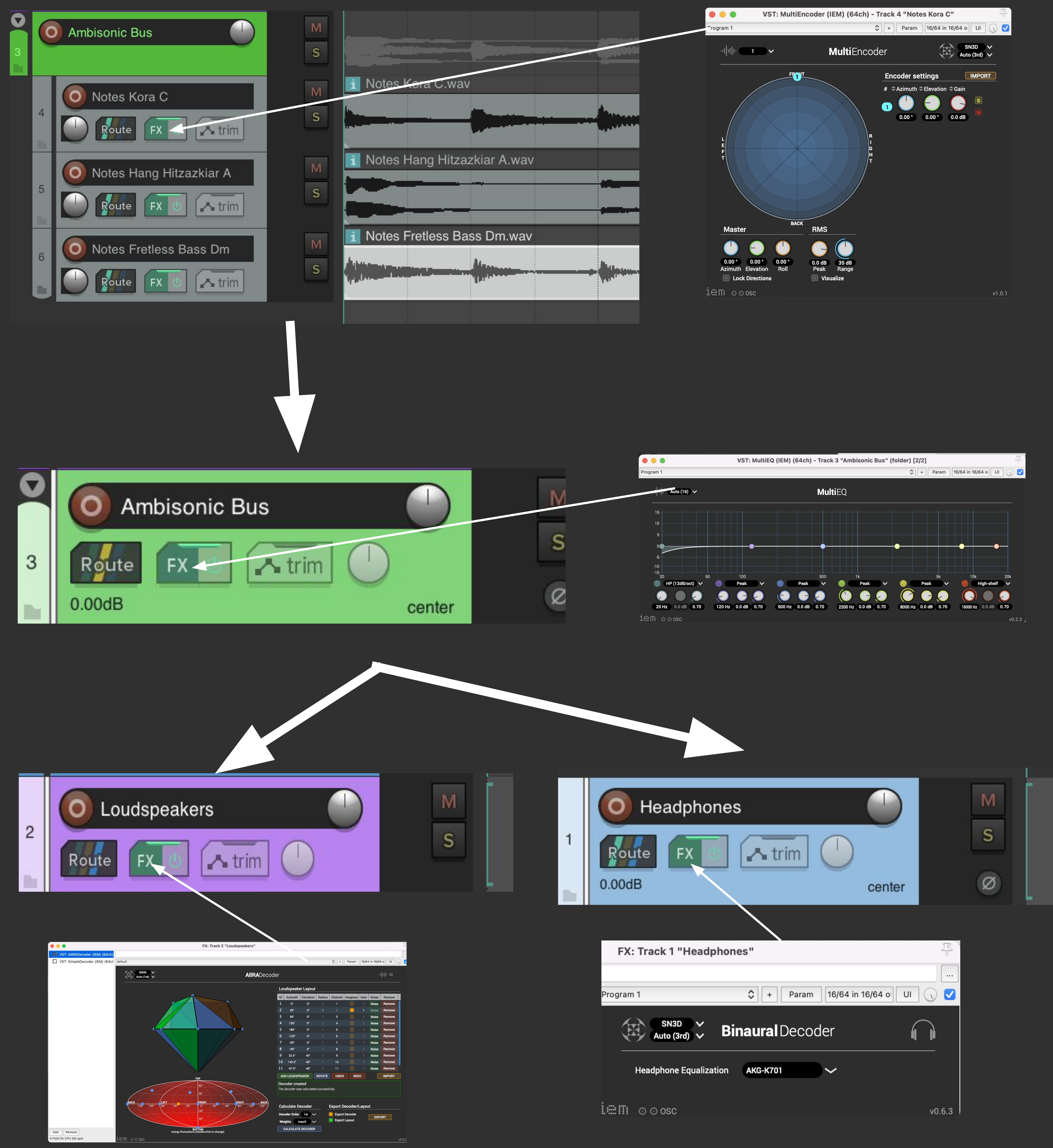
➔ **Number of channels pro Order**
Formula: $(\text{order} + 1)^2$
i.e. 4th order = 25 channels

# Let’s practice ambisonics!

# Open Reaper and the example project

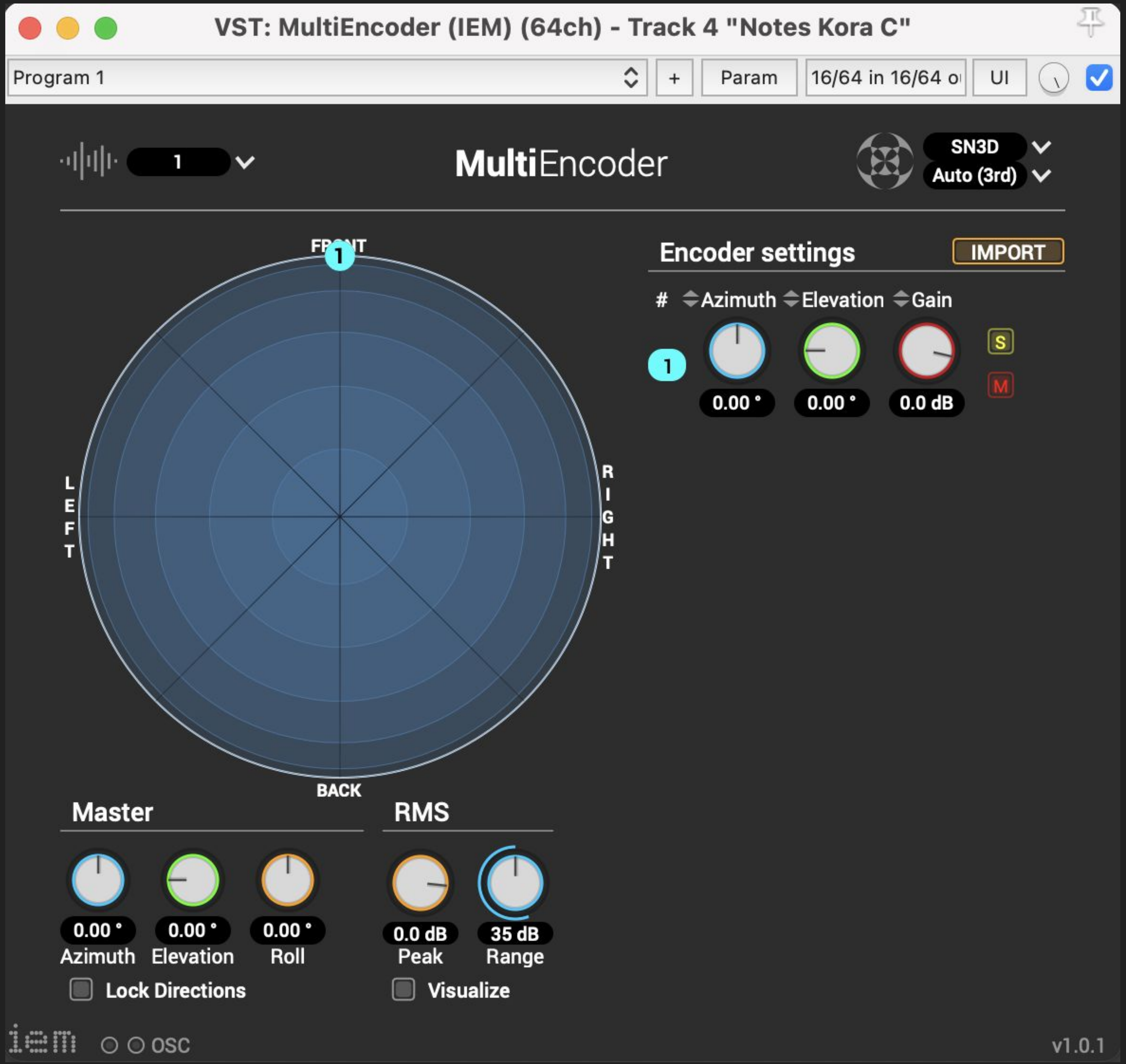
# Workflow Encoder-Decoder

- Add each of your source” tracks as child of the Ambisonics Bus (mono/stereo/multichannel/live)
- Each “source” track has its MultiEncoder VST as FX.
- The “Ambisonics Bus” track receives all encoders. Here you can add ambisonics effects (reverb, compression, EQ, etc). It mixes all ambisonics streams and can be used as a master track.
- The “Speakers Decoder” track receives audio channels from the “Ambisonics Bus” and decodes them to the actual speaker rig.
- The “Binaural Decoder” track receives from the “Ambisonics Bus” and sends a binaural decoded version to the headphones



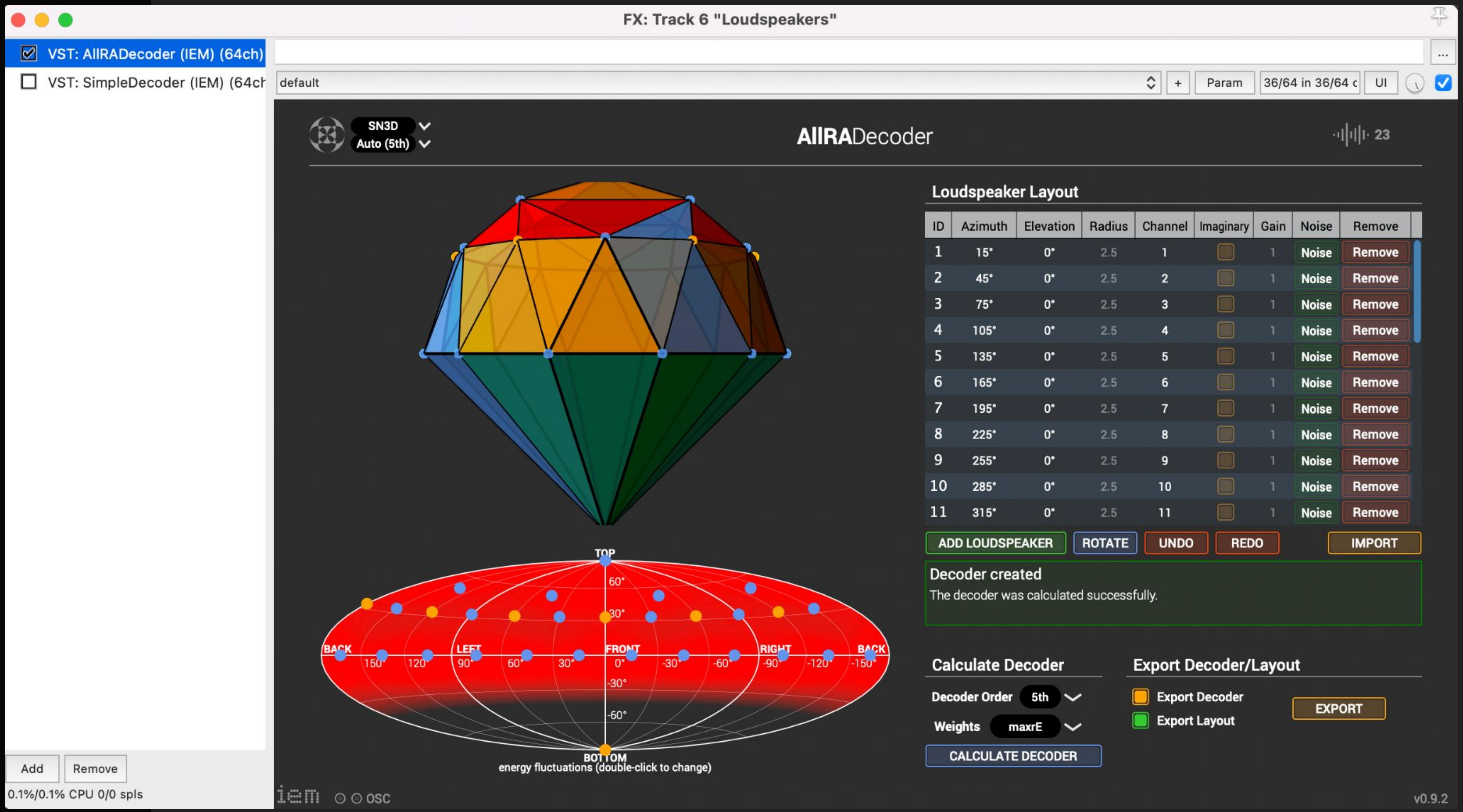
# Using the Encoder

- Select number of sound sources
- Select SN3D format and the ambisonics order



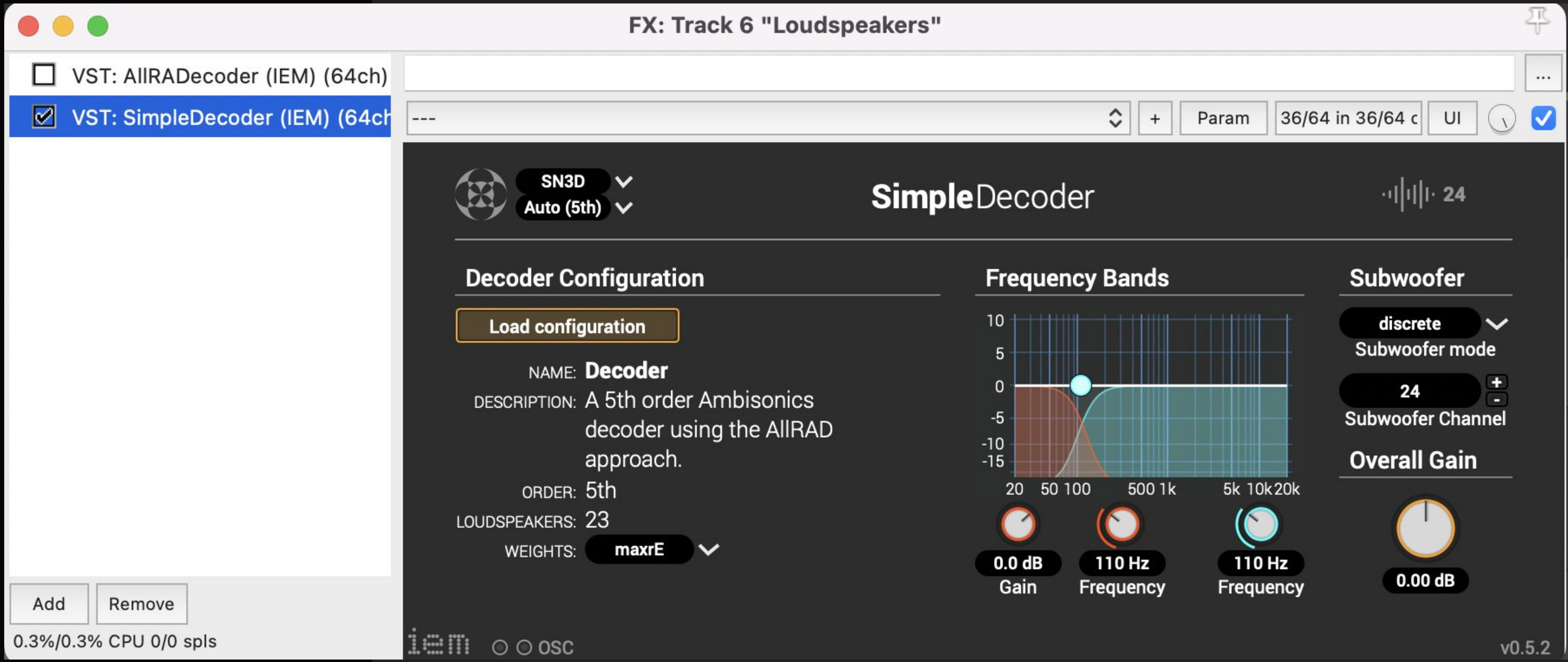
# Defining a decoder

- Select order
- Introduce your speaker positions
- Introduce “imaginary” speakers when needed for helping the decoder algorithm
- Export to save
- Import to load new ones



# Using the decoder

- We usually create decoders with AllRad VSTs but when using it, we prefer the SingleDecoder VST which incorporates subwoofer output and frequency band control.



# Using several types of sources: mono, stereo, multichannel, B-Format

- Practically, you will need to mix different types of sound sources, their encoded versions, and other ambisonic native audio files.

# TASK: AMBISONICS REAPER PROJECT

- download and install the IEM plugin suite
- download the reaper template
- open the binaural decoder and listen with headphones, change the positions of the source material or rotate the scene
- mute the BinauralDecoder, unmute the SimpleDecoder, connect to the speakers and try there
- Perceive and try to remember the difference between binaural and multichannel